

1. (1) TRUE

$$Y = X^T \beta_0 + \varepsilon$$

$$E(\varepsilon|X) = 0,$$

$$E(Y|X=x) = E(X^T \beta_0 + \varepsilon|X=x)$$

$$= \beta_0 E(X^T|X=x) + 0$$

$$= X^T \beta_0$$

(2) TRUE

$$E\{X(Y - X^T \beta_0)\} = 0$$

$$E\{X(X^T \beta_0 + \varepsilon - X^T \beta_0)\} = 0$$

$$E\{X\varepsilon\} = 0$$

$$E\{XY - XX^T \beta_0\} = 0$$

$$E(XY) - \beta_0 E(XX^T) = 0$$

$$\Rightarrow \beta_0 = \frac{E(XY)}{E(XX^T)} = \frac{E(XY)}{E(XX^T)^{-1} E(YY^T)}$$

(3) TRUE

$$E\{X(Y - X^T \beta_{proj})\} = 0$$

$$E\{X(X^T \beta_0 - X^T \beta_{proj} + \varepsilon)\} = 0$$

$$E\{XX^T \beta_0 - XX^T \beta_{proj} + X\varepsilon\} = 0$$

$$= E \left(XX^T \beta_0 - XX^T \beta_{proj} \right) - E(XE)$$

$$= E \left(XX^T \beta_0 \right) - E(XX^T) E(XX^T)^{-1} E(XY) - E(XE)$$

$$= \beta_0 E(XX^T) - E(XY) - E(XE)$$

$$= -E(XE)$$

$$= 0$$

④ TRUE

$$\frac{1}{n} \sum_{i=1}^n X_i (Y_i - X_i^T \beta) = 0 \text{ is}$$

sample version of the $E(X(Y - X^T \beta)) = 0$

As we are doing for β , sum of residuals weighted by X will be zero

⑤ TRUE

eq ④ describes OLS estimator. redefining normal equation

⑥ TRUE

$$E(Y|X) = X_S^T \gamma$$

conditional expectation of Y w.r.t X

~~for γ~~

here, if coefficients ~~are~~ for coordinates not in ~~zero~~ S have no effect on $E(Y|X)$

So they must be zero.

⑦ TRUE

$$\|Y - X(X^T X)^{-1} X^T Y\| \leq \|Y - X_S(X_S^T X_S)^{-1} X_S^T Y\|$$

regressing Y on X .

residuals from

regressing Y on X_S

So, left has full model it's a better fit so lower residual sum of squares

⑧ $\hat{Y} = X(X^T X)^{-1} X^T Y$

TRUE,

$$X_S(X_S^T X_S)^{-1} X_S^T Y = X_S(X_S^T X_S)^{-1} X_S^T \hat{Y}$$

fitted values from
 Y on X_S

$\hat{Y}$ on X_S

$\hat{Y}$ is projection of Y , so both will be same when projected on X_S .

9

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

OLS for X_S , $(X_S^T X_S)^{-1} X_S^T Y$.

for X , $(X^T X)^{-1} X^T Y$

$$(X_S^T X_S)^{-1} \neq (X^T X)^{-1}$$

So FALSE

10

TRUE

① $\hat{Y} = X(X^T X)^{-1} X^T Y$

TRUE, So, $\|Y - \hat{Y}\| = \|Y - X(X^T X)^{-1} X^T Y\|$

② $\hat{Y}_0 = X_S(X_S^T X_S)^{-1} X_S^T Y$

TRUE, So, $\|Y - \hat{Y}_0\| = \|Y - \hat{Y}_0\|$

$$= \|Y - X_S(X_S^T X_S)^{-1} X_S^T Y\|$$

③ $\hat{Y} - Y_0 = \hat{Y} - \hat{Y}_0$

$$= Y - \hat{Y}_0 - (Y - \hat{Y})$$

$$= \hat{Y} - Y_0$$

$$= \|\hat{Y} - Y_0\|$$

TRUE

(*)

(ii)

TRUE

true model only depends on x_s , coordinate, of β_0 not in S will be zero, then fitted & reduced models should be similar.

So, $\hat{y} - \hat{y}_s$ is very small,

4.

for x , t -value ???

P-value ???

$$t\text{-value} = \frac{\text{Estimate}}{\text{Standard error}}$$

	estimate	S.E	t-value	Pr(> t)
(intercept)	1.0983	0.1589	6.912	1.63e-07
x	0.7190	0.1535	???	???

$$t\text{-Value} = \frac{0.7190}{0.1535} = 4.684$$

$$\text{for } x, P(T > |t|) = \frac{\widehat{\text{Coeff of } x}}{\text{SE}(\widehat{\text{Coeff of } x})} = \frac{0.7190}{0.1535} = 4.683$$

$$P(T > 4.683) = 2 \times P(T > |t|)$$

$$\text{prob, } P(T > |t|) = 1 - pt(4.683, 28)$$

$$= \cancel{0.0002465} \quad \left(\begin{array}{l} \text{calculated} \\ \text{on R, refer} \\ \text{code} \end{array} \right)$$

$$P(T > 4.683) = 2 \times \cancel{0.0002465}$$

$$= \cancel{4.93 \times 10^{-5}} = 6.59 \times 10^{-5}$$

Multiple R-squared ???

$$R^2 = \frac{F \times df_1}{F \times df_1 + df_2} = \frac{21.94}{21.94 + 28}$$

$$\begin{aligned} F_{\text{static}}, F &= 21.94 &= \frac{21.94}{49.94} \\ df_1 &= 1 \\ df_2 &= 28 &\approx 0.439 \end{aligned}$$